

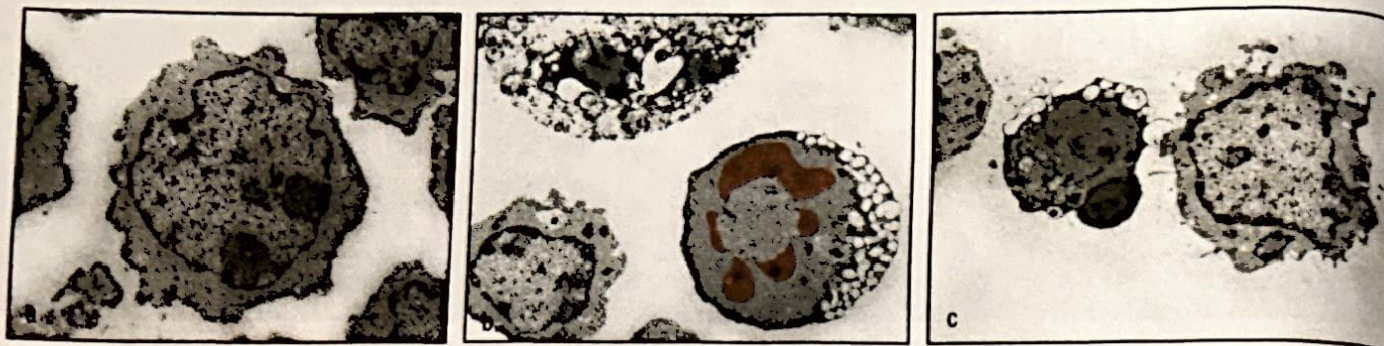


Fig. 19.1 Apoptosis. Apoptosis is a form of induced 'cell suicide' in which the cell undergoes chromatin compaction and DNA fragmentation, followed by cell shrinkage and internal degradation. Panel (a) shows a healthy cell with a normal nucleus. Early in apoptosis (panel b), the chromatin in the nucleus becomes condensed (red) and, although the cell sheds membrane vesicles, the integrity of the cell membrane is retained, in contrast

to the necrotic cell in the upper part of the same field. In late stages of apoptosis (panel c), the cell nucleus (middle cell) is very condensed, no mitochondria are visible, and the cell has lost much of its cytoplasm and membrane through the shedding of vesicles. Photographs ($\times 3500$) courtesy of R. Windsor and E. Hirst.

The case of Ellen O'Hara: uncontrolled lymphocyte proliferation in the absence of infection or malignancy.

18-month-old girl, enlarged spleen. Order blood tests.

Ellen O'Hara was born after a normal and uncomplicated pregnancy, was breast fed, and received her routine immunizations without any adverse reactions. At 18 months old, during a routine check-up by her pediatrician, she was found to have an enlarged spleen (splenomegaly) and extensive enlargement of her lymph nodes (lymphadenopathy) (Fig. 19.3). According to her parents, she had had no unusual infections and seemed to be growing and developing normally.

Laboratory tests revealed that Ellen's white blood cell count was $12,500 \mu\text{L}^{-1}$, of which 9175 were lymphocytes (normal 3000–7500). Her serum immunoglobulins

Fig. 19.2 Binding of FASLG to FAS initiates the process of apoptosis in the FAS-bearing cell. Binding of trimeric FASLG to trimeric FAS brings the death domains in the FAS cytoplasmic tails together. A number of adaptor proteins containing death domains bind to the death domains of FAS, in particular the protein FADD, which in turn interacts through a second death domain with the protease caspase 8. Clustered caspase 8 can transactivate, cleaving caspase 8 itself to release an active caspase domain that in turn can activate other caspases. The ensuing caspase cascade culminates in activation of the caspase-activatable DNase (CAD), which is present in all cells in an inactive cytoplasmic form bound to an inhibitory protein called I-CAD. When I-CAD is broken down by caspases, CAD can enter the nucleus, where it cleaves DNA into the 200-bp fragments that are characteristic of apoptosis.

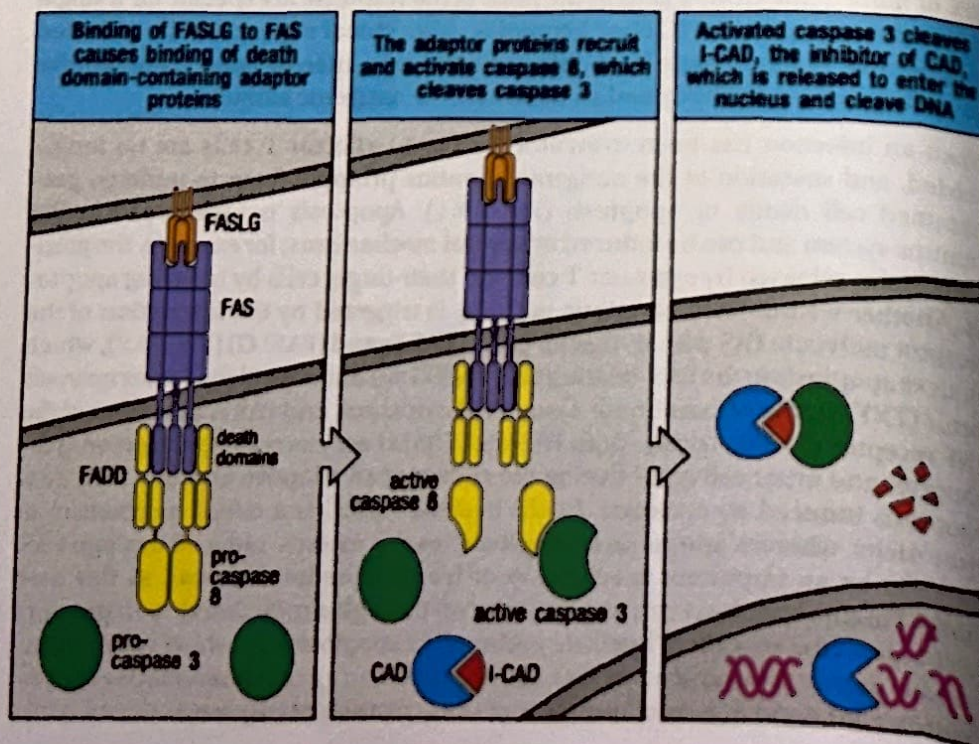




Fig. 19.3 Lymphadenopathy in ALPS. Young girl with ALPS with very enlarged lymph nodes in her neck.

were all elevated: IgG, 4000 mg dL⁻¹ (normal 520–1500); IgM, 400 mg dL⁻¹ (normal 40–200); and IgA, 1660 mg dL⁻¹ (normal 27–169). Flow cytometry analysis of her lymphocytes revealed that 29% were CD19⁺ B cells (normal 5–15%; CD19 is a component of the B-cell co-receptor complex) and 65% were CD3⁺ T cells (normal 61–84%; CD3 is a component of the T-cell receptor complex). Of the CD3⁺ T cells, 14% carried the co-receptor protein CD4 and 18% the co-receptor CD8. She thus had many CD3⁺4⁺8⁺ T cells. Of these, the vast majority expressed TCRαβ (the αβ form of the T-cell-receptors) and hence were TCRαβ⁺ double-negative (DN) T cells (normally, TCRαβ⁺ DN T cells are either absent or constitute less than 2.5% of circulating T cells). A biopsy of a lymph node from Ellen's neck showed extensive enlargement of the follicles (hyperplasia) and a marked increase in the numbers of immunoblasts and plasma cells in the paracortical area. No infectious agents were cultured from the lymph node, despite the fact that the observed changes resembled those caused by a viral infection. Although more than 50% of the T cells in the lymph node were double negatives, no chromosomal abnormality was found on karyotyping, and there was no evidence of oligoclonality of the T-cell receptor, thus ruling out a malignancy.

In the absence of evidence of infection or malignant disease, autoimmune lymphoproliferative disease was diagnosed and Ellen received the anti-inflammatory steroid prednisone and the immunosuppressant drug cyclosporin A. Her lymph nodes rapidly reduced in size after this therapy, but enlarged again when therapy was discontinued.

Ellen continued to grow and develop normally, and when she reached adolescence the size of her lymph nodes decreased spontaneously. At 18 years of age, repeat blood counts revealed that her platelet count was 75,000 μL⁻¹ (normal 150,000–250,000). An autoantibody against platelets was found in her serum. A diagnosis of idiopathic thrombocytopenic purpura (low platelet numbers accompanied by red or purplish-red spotty skin discoloration due to local hemorrhages) was made. She was treated with the steroid dexamethasone, and the condition resolved. At age 32, Ellen's blood neutrophil count fell to <1000 μL⁻¹ (normal 2500–5000). She was found to have developed an autoantibody against granulocytes.

Ellen's family history was informative: her paternal grandfather had splenomegaly and generalized lymphadenopathy as a child, and his spleen was removed at

Increased B cells, large number of DN T cells.

No evidence of infection or malignancy. ALPS?

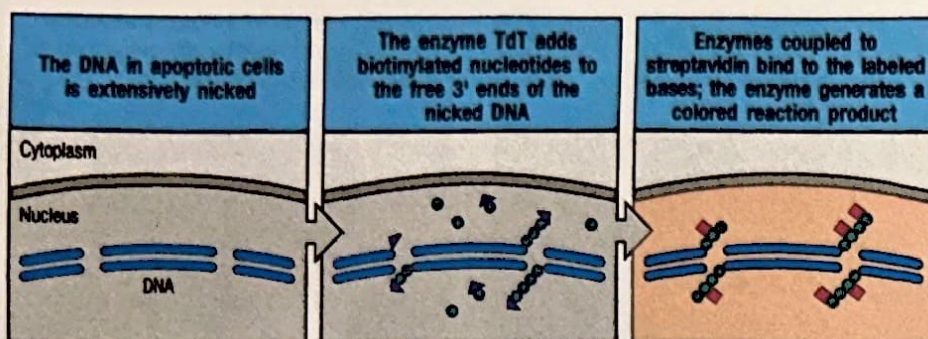


Fig. 19.4 The TUNEL assay. When cells undergo apoptosis, their DNA becomes fragmented (first panel). By labeling the fragmented DNA with biotin-labeled nucleotides attached by the enzyme terminal deoxynucleotidyltransferase (TdT), the apoptotic cells can be clearly seen under light microscopy. TdT adds nucleotides to the ends of DNA fragments; biotin-labeled dUTP is most commonly added in this assay (second panel). The biotinylated

DNA can be detected by using streptavidin, which binds to biotin, coupled to enzymes that convert a colorless substrate into a colored insoluble product (third panel). Cells stained in this way can be detected by light microscopy, as shown in the photograph of apoptotic cells (stained red) in the thymic cortex. Photograph courtesy of R. Budd and J. Russell.

TUNEL assay,
defective apoptosis.

age 25. At age 60, he developed a B-cell lymphoma. Ellen's father also had splenomegaly and lymphadenopathy but no clinical symptoms. When blood lymphocytes from Ellen's father and paternal grandfather were examined by flow cytometry, a large number of double-negative T cells were found. In contrast, her brother, mother, and maternal grandparents had normal T cells. The TUNEL assay for apoptotic cells (Fig. 19.4) was performed on blood mononuclear cells from Ellen, her parents, and her paternal grandfather. The cells were first stimulated *in vitro* with phytohemagglutinin for 3 days, and growth of the resulting T-cell blasts was continued for 3 weeks by the addition of IL-2 to the cultures. The cultures were then divided; half were exposed to an antibody to FAS, which mimics the function of FASLG. The percentage of cells undergoing apoptosis was then counted. Sixty percent of her mother's T cells underwent apoptosis, whereas only 2% of Ellen's cells, <1% of her father's cells, and 1.4% of her paternal grandfather's cells demonstrated programmed cell death (normal controls 35–70%). The *FAS* and *FASLG* genes were examined in DNA samples from Ellen, her father, and her paternal grandfather. An identical single-base transversion, causing a premature termination codon, was found in one of the alleles of the *FAS* gene in these DNA samples. The *FAS* genes in Ellen's mother and brother were normal.

Autoimmune lymphoproliferative syndrome (ALPS).

ALPS is characterized by splenomegaly and lymphadenopathy from early childhood, and, frequently, autoimmunity. Affected individuals can develop autoimmune hemolytic anemia, neutropenia, and thrombocytopenia, and are at increased risk of developing lymphoma. Most patients with ALPS are heterozygous for a dominant mutation in the *FAS* gene, and their activated T cells do not undergo FAS-mediated apoptosis *in vitro*, as is the case for Ellen, her father, and his father. This form of ALPS is also known as ALPS-FAS. Patients with ALPS due to *FAS* mutations also have elevated serum levels of FASLG, IL-10, IL-18, and vitamin B₁₂, which can be used as reliable biomarkers, along with the increase in DN T lymphocytes. In some cases, ALPS is due to somatic mutations of *FAS* that occur in an early lymphoid progenitor. Because of the impairment of apoptosis, the proportion

of lymphocytes carrying the somatic mutations may increase over time, and is particularly high among DN T cells.

Other patients with ALPS have been found to have mutations in the genes encoding FASLG (ALPS-FASLG) or caspase 10 (ALPS-CASP10), an enzyme involved in triggering apoptosis via the FAS pathway (ALPS2A). However, other mechanisms, not related to defects of the FAS signaling pathway, may also cause ALPS.

In particular, bi-allelic loss-of-function mutations of the protein kinase C δ (*PRKCD*) gene may also cause ALPS. Protein kinase C δ controls growth and apoptosis in multiple cell types, including T and B lymphocytes. *PRKCD* loss-of-function mutations results in impairment of B cell tolerance. Although serum levels of immunoglobulins in patients with *PRKCD* deficiency may be variable, autoantibody production is a common finding. As in other forms of ALPS, patients manifest lymphadenopathy and hepatosplenomegaly; however, the number of TCR $\alpha\beta$ ⁺ double-negative cells is not increased. The proportion of immature (transitional) B cells is markedly increased, whereas the number of memory B cells is decreased.

ALPS is also caused by gain-of-function somatic mutations of the *NRAS* or *KRAS* genes that impair IL-2 withdrawal-induced apoptosis by suppressing the proapoptotic protein Bim, which controls mitochondrial stability upon cytokine deprivation.

Another cause of ALPS is represented by heterozygous loss of function mutations of *CTLA4*. This molecule is expressed by T cells and has a higher affinity than CD28 for binding to CD80 and CD86 molecules. Upon binding to these molecules, *CTLA4* delivers an inhibitory signal to T lymphocytes. *CTLA4* deficiency is characterized by autoimmune cytopenias, hypogammaglobulinemia, and lymphocytic infiltrates in multiple organs, especially the gastrointestinal tract, the lungs and the brain. Other possible manifestations of immune dysregulation include psoriasis, autoimmune thyroiditis, and arthritis. *CTLA4* plays an important role in regulatory T cell (*T_{reg}*) function. Patients with *CTLA4* deficiency show an abnormal *T_{reg}* phenotype, with reduced expression of FOXP3 and CD25, and reduced function.

Finally, mutations of the FAS-associated protein with death domain (FADD), an adaptor molecule downstream of FAS (Fig. 19.2) cause autoimmune lymphoproliferation associated with liver disease, encephalopathy, and cardiac malformations.

Treatment of ALPS is mostly based on immune suppression and the surveillance of tumors. Splenectomy should be avoided, because of the risk of infections by encapsulated bacteria (see Case 30). Immunosuppression with CTLA4-Ig (abatacept) or with sirolimus is beneficial in patients with *CTLA4* deficiency.

The clinical and immunologic features of ALPS bear a striking resemblance to a lymphoproliferative disease observed in mice with *lpr* or *gld* mutations. The *lpr* phenotype results from the absence of Fas, whereas the *gld* phenotype is caused by a mutation in *Faslg*. A progressive accumulation of DN T cells is observed in both these strains of mice (note that these circulating CD3⁺ DN T cells should not be confused with the immature CD3⁺4⁺8⁻ 'double-negative' thymocytes that are a normal stage of T-cell development in the thymus). The mice make antibodies against double-stranded DNA, similar to the situation in human systemic lupus erythematosus (see Case 36). Consistent with these findings in mice, patients with ALPS due to defects of FAS or of the FAS signaling pathway have defective T-cell apoptosis and abnormal accumulations of DN T cells. When B cells are activated, they also express FAS and become susceptible to FAS-mediated apoptosis. Thus, activated B cells in ALPS are not properly eliminated. The serum concentrations of immunoglobulins is increased (hypergammaglobulinemia), and so is the number of B cells (B-cell lymphocytosis), and pathological autoantibody production ensues. Because T cells and B cells are not eliminated normally, patients with ALPS are predisposed to develop lymphomas. Autoimmunity may result because FAS-mediated killing is a mechanism for removing autoreactive B cells.

Questions.

- 1 Most patients with ALPS are heterozygous for the mutation in FAS; they have one normal allele and one mutant allele. How do you explain the dominant inheritance?
- 2 Ellen's great-aunt (her paternal grandfather's sister) was found to have the same FAS mutation as Ellen, yet she had no symptoms. How can this be explained?
- 3 It is advantageous for viruses to inhibit apoptosis so that the host cells in which they thrive do not get eliminated by apoptosis induced by recognition by cytotoxic T cells. How might a virus accomplish this?
- 4 When FAS is activated by FASLG, it associates with and activates caspase 8 (see Fig. 19.2). When the gene encoding caspase 8 is knocked out in mice, this proves to be lethal at the fetal stage. Would it be worthwhile to search for caspase 8 mutations in patients with ALPS when there is no mutation in FAS or FASLG?

CASE 20

HYPER IgE SYNDROME

A defect in T_H17 lineage differentiation.

After antigen recognition and concomitant activation by specific sets of cytokines, naive $CD4^+$ T cells differentiate into distinct lineages of helper or regulatory effector T cells, characterized by distinct biologic functions and cytokine expression profiles. About ten years ago, T_H17 cells were identified as a distinct lineage of differentiated $CD4$ T cells, characterized by expression of the cytokines IL-17A (IL-17), IL-17F, IL-21, and, in some cases, IL-22 (Fig. 20.1). IL-17 and IL-17F have similar activities that include the induction in epithelial cells of both chemokines that attract neutrophils and β -defensins (antimicrobial peptides). T_H17 cells play a role in the immune response to extracellular bacteria and fungi, and in autoimmune diseases.

In humans, IL-6, in synergy with IL-1 β , IL-21, and possibly TGF- β , has a key role in the differentiation of T_H17 cells. IL-23 is important in amplifying the generation of T_H17 cells and sustaining their survival. IL-6, IL-21, and IL-23 all bind to the same class of heterodimeric receptors, which, upon binding, activate Janus kinases (JAKs) that phosphorylate the intracellular portions of the receptor. The inactive form of the transcription factor STAT3 is recruited to the phosphorylated receptor and is itself phosphorylated (Fig. 20.2). Phosphorylated STAT3 dissociates from the receptor and forms homodimers that translocate to the nucleus and act as transcription factors to induce expression of the retinoic acid-related orphan receptor γT (ROR γT). The differentiation of activated $CD4^+$ T cells into T_H17 cells is determined by the expression of the transcription factor ROR γT .

TOPICS BEARING ON THIS CASE:

Neutrophil function

T-cell subsets and their differentiation

T_H17 cell function

IgE class antibody